DIFFERENTIAL EQUATION

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INTRODUCTORY CONCEPTS

DEFINITION

The laws of the universe are written largely in the language of mathematics. Many important and significant problems in engineering, the physical sciences and the social sciences when formulated in mathematical terms require the determination of a function satisfying an equation containing the derivatives of unknown function. Such equations are called **Differential equation**.

Differential Equation(DE) is an equation involving an unknown function and one or more of its derivatives. The study of DE has two principal goal:

a. To discover DE that describes a physical situation.

b. To find the appropriate solution of that equation

EXAMPLE

The following are some general examples of differential equations;

1. $\frac{dy}{dx} = \cos x$
2. $\frac{d^2y}{dx^2} + ky^2 = 0$
3. $(x^2 + y^2)dx - 2xydy = x$
4. $(\frac{d^2w}{dx^2})^3 - xw\frac{dw}{dx} + w = 0$

When an equation involves one or more derivatives with respect to a particular variable, that variable is called an **Independent variable**. A variable is called **Dependent variable** if a derivative of that variable occurs. In the equation:

$$\frac{d^2v}{dx^2} + \frac{dv}{dx} = 0$$

has one dependent variable **v** and an independent variable **x**.

FORMS OF DIFFERENTIAL EQUATION

If the equation contains ordinary derivatives only, it is called **Ordinary Differential Equation(ODE)**. If it contains partial derivatives, it is a **Partial Differential Equation(PDE)**.

In this semester's work,our attention would be focused on ODEs.

An ODE can be written as:

$$F(x, y, y', y'', \cdots, y^n) = 0$$

ORDER AND DEGREE

The **order** of a differential equation is the order of the highest derivative that appears in the equation.

Eg.

$$\frac{dy}{dx} - 5y = 0$$

is an ODE with **order 1**

The **degree** of the differential equation is the power or index that the highest derivative is raised if the equation is rationalized and cleared of fractions with regard to all derivatives involved in it.

Eg.

$$[1 + (\frac{dy}{dx})^2]^3 = k^2(\frac{d^2y}{dx^2})^2$$

has a **degree** 2

LINEAR DIFFERENTIAL EQUATION AND SOLUTION OF A DIFFERENTIAL EQUATION

LINEAR DIFFERENTIAL EQUATION

A differential equation is linear if it can be put in the form:

$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_2(x)y'' + a_1(x)y' + a_0(x)y = f(x)$

where the subscripted a's are functions of x.

In other words, a differential equation is said to be linear if:

(i) The dependent variable and all its derivatives only occurs to the first power.

(ii) There are no terms involving products of the dependent variable and the derivatives of the dependent variable.

(iii) There are no functions of the dependent variable and its derivatives which are in trigonometric, logarithmic or exponential form.

EXAMPLES OF LINEAR D.E

1. $\frac{dy}{dx} + y = x^2$
2. $y^{''} + y^{'} = 0$
3. $6y^{'} + 5y = \cos x$

EXAMPLES OF NON-LINEAR D.E

1. $\frac{dy}{dx} + y^2 = 0$
2. $(\frac{dy}{dx})^2 + 3y = 0$

SOLUTION OF A DIFFERENTIAL EQUATION

DEFINITION

A solution of a differential equation is any function, say ϕ, satisfying the given differential equation. A solution may be defined on the whole real line or on only a part of the line often an interval.

Thus if ϕ is a solution of some first order differential equation, say

$$F(x, y, \frac{dy}{dx}) = 0$$

on an interval I(real), then it implies

$$F(x, \phi, \frac{d\phi}{dx}) = 0$$

EXAMPLE

Show that for any values of the arbitrary constant c_1 and c_2 the function

$$\phi = c_1 \cos x + c_2 \sin x$$

is a solution of the differential equation $\frac{d^2y}{dx^2} + y = 0$

Solution

If ϕ is a solution to $F(y, \frac{d^2y}{dx^2}) = 0$, then $F(\phi, \frac{d^2\phi}{dx^2}) = 0$. If ϕ proves otherwise then it is not a solution to the differential equation.

We differentiate ϕ twice to obtain $\frac{d^2\phi}{dx^2}$

$$\frac{d\phi}{dx} = -c_1 \sin x + c_2 \cos x$$

$$\frac{d^2\phi}{dx^2} = -c_1 \cos x - c_2 \sin x$$

$$\Rightarrow \frac{d^2\phi}{dx^2} + \phi = (-c_1 \cos x - c_2 \sin x) + (c_1 \cos x + c_2 \sin x) = 0$$

This implies we have an identity. Hence ϕ is a solution to the given ODE.

EXERCISE

Prove that each of the following equations is a solution to the differential equation given beside it.

1. $y = \frac{x^3}{4} + \frac{c}{x}$ $\qquad x\frac{dy}{dx} + y = x^3$
2. $y = A\sin t + B\cos t + t$ $\qquad \frac{d^2y}{dt^2} + y = t$
3. $ye^x = x + c$ $\qquad \frac{dy}{dx} + y = e^{-x}$

THE ELIMINATION OF ARBITRARY CONSTANT

Methods for the elimination of arbitrary constant vary with the way in which the constants enter the relation. A method, which is efficient for one problem may be poor for another. One fact persists throughout, that the number of derivatives that need be used is the same as the number of arbitrary constants to be eliminated.

We shall in each case determine the differential equation that is:

(a) Of order equal to the number of arbitrary constants in the given relation.

(b) Consistent with that relation.

(c) Free from arbitrary constants.

EXAMPLE

Eliminating the arbitrary constants c_1 and c_2 from the relation:

$$y = c_1 e^{-2x} + c_2 e^{3x} \cdots (1)$$

Since two constants are to be eliminated, obtain the two derivatives:

$$y' = -2c_1 e^{-2x} + 3c_2 e^{3x} \cdots (2)$$

$$y'' = 4c_1 e^{-2x} + 9c_2 e^{3x} \cdots (3)$$

The elimination of c_1 from(2) and (3) yields:

$$y'' + 2y' = 15c_2 e^{3x}$$

The elimination of c_1 from(1) and (2) yields:

$$y' + 2y = 5c_2 e^{3x}$$

Hence

$$y'' + 2y' = 3(y' + 2y)$$

or

$$y'' - y' - 6y = 0$$

It is clear that the most general solution of an ODE is in the unknowns always involves a number of independent arbitrary constants equal to the ODE.

GENERAL SOLUTION

Definition

A formula that gives all solutions of a differential equations is called the general solution of the equation.

EXAMPLE

Find the differential equation whose general solution is $y = c\cos x$

SOLUTION

$$y = c\cos x \cdots (1)$$

Since it has one constant we expect its differential equation to be 1st order then:

$$\frac{dy}{dx} = -c\sin x \cdots (2)$$

From (1) and (2):

$$c = \frac{y}{\cos x} = -\frac{\frac{dy}{dx}}{\sin x}$$

$$\Rightarrow -\cos x\frac{dy}{dx} = y\sin x$$

$$y'\cos x + y\sin x = 0$$

Alternatively, from (1): $c = \frac{y}{\cos x}$

Differentiating w.r.t. x: $0 = \frac{\cos x\frac{dy}{dx} + y\sin x}{\cos^2 x}$

$$\Rightarrow y'\cos x + y\sin x = 0$$

SUBSIDIARY CONDITIONS

Subsidiary Condition(s) is/are condition(s) or set of conditions, on the differential equation that will allow us to determine which solution that we are after.

INITIAL-VALUE PROBLEM

Initial Value Problem is the differential equation along with subsidiary conditions on the unknown function and its derivatives, all given at the same value of the independent variable.

BOUNDARY-VALUE PROBLEM

Boundary Value Problem is the differential equation along with subsidiary conditions on the unknown function and its derivatives, which are given at more than one value of the independent variable.

If initial condition(s) is/are used to solve a differential equation, the solution is then called **particular solution**

ORDINARY DIFFERENTIAL EQUATION OF FIRST ORDER

FIRST ORDER DIFFERENTIAL EQUATION

First order differential equations is an equation containing only the first differential. In its general form, the 1st order differential equation is given by:

$$F(x, y, \frac{dy}{dx}) = 0 \quad or \quad F(x, y, y') = 0$$

The above equation has **x** called the **Independent variable**

y as the **dependent variable**

y' the first derivative of y w.r.t x

EXAMPLES

1. $\frac{dy}{dx} - 5x = 0$
2. $y' - 4xy = x^3 e^x$
3. $x\frac{dy}{dx} - y = 0$
4. $3y' = 4xy$

STANDARD FORM

Standard form for a first order differential equation in the unknown function y is:

$$y' = f(x, y)$$

DIFFERENTIAL FORM

The right hand side of the standard form [f(x,y)] can be written as a quotient of two other functions **M(x,y)** and **-N(x,y)**. Then $\frac{dy}{dx} = \frac{M(x,y)}{-N(x,y)}$ which is equivalent to the Differential form.

$$M(x, y)dx + N(x, y)dy = 0$$

GENERAL AND PARTICULAR SOLUTIONS OF A FIRST ORDER ORDINARY DIFFERENTIAL EQUATION

In this section, we examine methods of solving a first order differential equation. The methods to consider are as follows;

1. Method of Separation of Variables
2. Homogenous
3. Method of Integrating factor used for linear first order ordinary differential equation and exact (non-exact) differential equations
4. Bernoulli Equations
5. Riccati Equations

SEPARABLE EQUATIONS

The general form is

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

If this is separable, then

$$M(x,y) = g(x)s(y)\cdots(i)$$

and

$$N(x,y) = h(x)r(y)\cdots(ii)$$

If $M(x,y)$ and $N(x,y)$ can be expressed in the form in (i) and (ii) respectively then is called **Separable Differential equation**.

For this separable differential equation, we divide throughout by $s(y)h(x)$

$$g(x)s(y) + h(x)r(y)\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{g(x)}{h(x)}dx + \frac{r(y)}{s(y)}dy = 0$$

Whose solution is;

$$\int \frac{g(x)}{h(x)}dx + \int \frac{r(y)}{s(y)}dy = c$$

EXAMPLE

Solve the equation(IVP)

$$(x+1)y + x(y-1)\frac{dy}{dx} = 0 \qquad y(1) = 1$$

SOLUTION

$$(x+1)y + x(y-1)\frac{dy}{dx} = 0$$

Divide throughout by xy

$$\frac{x+1}{x} + \frac{y-1}{y}\frac{dy}{dx} = 0$$

$$\Rightarrow \int \frac{x+1}{x}dx + \int \frac{y-1}{y}dy = c$$

$$x + \ln|x| + y - \ln|y| = c$$

$$1 + \ln|1| + 1 - \ln|1| = c \quad \Rightarrow c = 2$$

$$\Rightarrow x + \ln|x| + y - \ln|y| = 2$$

$$\Rightarrow x + y - 2 = \ln(\frac{y}{x})$$

$$\Rightarrow \frac{y}{x} = e^{x+y-2}$$

HOMOGENEOUS EQUATIONS

A function of two variables, f(x,y), is said to be **homogeneous of degree n** if there is a constant $n \in R$ such that:

$$f(\lambda x, \lambda y) = \lambda^n f(x, y)$$

for all λ, x and y for which both sides are defined. A differential equation of the same degree. To solve homogeneous equations, turn them into separable ones using the substitution:

$$y = xv$$

where v is a function of x; also:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

EXAMPLE

Solve the differential equation

$$(x^2 + y^2)dx - 2xydy = 0$$

SOLUTION

Notice that the function $M(x, y) = x^2 + y^2$ and $N(x, y) = -2xy$ are both homogeneous.

The substitution y=xv

implies that dy=xdv +vdx, and given equation is transformed into:

$$\frac{1}{x}dx = \frac{2v}{1 - v^2}dv$$

Integrating both sides gives:

$$log|x| = -log|1 - v^2| + log|c|$$

$$log|x| = log|\frac{c}{1-v^2}|$$

Finally, to write the solution in terms of the original variables, x and y, we replace v by $\frac{y}{x}$

$$x = \frac{c}{1-(\frac{y}{x})^2}$$

$$x^2 - y^2 = cx$$

EQUATION WITH LINEAR COEFFICIENTS

These equations are of theform:

$$(ax + by + c) + (fx + gy + h)y' = 0$$

where a,b,c,f,g,h are constants where

$$M(x, y) = ax + by + c \text{ and } N(x, y) = fx + gy + h$$

written as $M(x, y) + N(x, y)y' = 0$, which are not homogeneous. However the differential equation can be made homogeneous and solved accordingly provided

$$\triangle = \begin{vmatrix} a & b \\ f & g \end{vmatrix} \neq 0$$

If $\triangle = 0$, another technique can be used to solve the differential equation.

if $\triangle \neq 0$, solve

$$ax + by + c = 0$$

$$fx + gy + h = 0$$

simultaneously to obtain $x = x_0$ and $y = y_0$.
Then substitute $X = x - x_0$ and $Y = y - y_0$ to convert the differential equation to a homogeneous differential equation, which takes the form:

$$(aX + bY) + (fX + gY)Y' = 0$$

Then by the substitution: Y=vX
Y' = v+v'X, we can solve the original differential equation.

EXAMPLE (METHOD 1)

Solve the differential equation:

$$2x + 9y - 20 - (6x + 2y - 10)y' = 0$$

SOLUTION

$$\triangle = \begin{vmatrix} 2 & 9 \\ -6 & -2 \end{vmatrix} = 50 \neq 0$$

Hence, we then solve

$$2x + 9y = 20$$

$$6x + 2y = 10$$

simultaneously to obtain $x_0 = 1, y_0 = 2$

Then let $X = x - 1$ and $Y = y - 2$

$$\Rightarrow [2(X+1)+9(Y+2)-20]-[6(X+1)+2(Y+2)-10]Y'=0$$

$$\Rightarrow [2X+9Y]-[6X+2Y]Y'=0$$

Substituting

$$Y=vX \Rightarrow Y'=v+v'X$$

$$\Rightarrow [2X+9vX]-[6X+2vX][v+v'X]=0$$

$$\Rightarrow (2+9v)-(6+2v)(v+v'X)=0$$

$$\Rightarrow 2+9v-v(6+2v)-X(6+2v)v'=0$$

$$\Rightarrow (2+3v-2v^2) - X(6+2v)v' = 0$$

but $v' = \frac{dv}{dX}$

$$\Rightarrow (\frac{6+2v}{2+3v-2v^2})dv = \frac{dX}{X} \Rightarrow \frac{6+2v}{(2-v)(1+2v)}dv = \frac{dX}{X}$$

Using partial fraction, we have:

$$\Rightarrow \int [(\frac{2}{2-v}) + (\frac{2}{1+2v})]dv = \int \frac{dX}{X}$$

$$\Rightarrow -2\ln|2-v| + \ln|1+2v| = \ln|x| + \ln c$$

$$\Rightarrow -\ln c = \ln |X\frac{(2-v)^2}{1+2v}|$$

$$\Rightarrow \frac{X(2-v)^2}{1+2v} = \frac{1}{c} = A$$

Since $v = \frac{Y}{X}, \Rightarrow \frac{X(2-\frac{Y}{X})^2}{1+2\frac{Y}{X}} = A \Rightarrow \frac{(2X-Y)^2}{X+2Y} = A$

$$\Rightarrow [2(x-1)-(y-2)]^2 = A[(x-1)+2(y-2)]$$

$$\Rightarrow [2(x-1)-(y-2)]^2 = A(x+2y-5)$$

$$(2x-y)^2 = A(x+2y-5)$$

EXAMPLE 2 (METHOD 2)

Solve the differential equation $(x+2y-3)+(2x+4y+5)y'=0$

SOLUTION

$$\triangle = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$$

Take $z = x+2y \Rightarrow z' = 1+2y' \qquad y' = \frac{z'-1}{2}$

The differential equation becomes:

$$(z-3)+(2z+5)(\frac{z'-1}{2}) = 0$$

$$\Rightarrow 2z-6+(2z+5)(z'-1) = 0$$

$$\Rightarrow 2z-6-(2z+5)+(2z+5)z' = 0$$

$$\Rightarrow -11+(2z+5)z'$$

Separating variables, we have:

$$\int 11dx = \int (2z + 5)dz$$

$$\Rightarrow 11x + c = z^2 + 5z, \text{ but } z = x + 2y$$

$$\Rightarrow 11x + c = (x + 2y)^2 + 5(x + 2y)$$

$$\Rightarrow 11x + c = (x + 2y)^2 + 5x + 10y$$

$$\Rightarrow (x + 2y)^2 - 6x + 10y = c$$

EXACT EQUATION

Given a function f(x,y), its total differentials, df, is defined as:

$$df = \frac{\delta f}{\delta x}dx + \frac{\delta f}{\delta y}dy$$

This shows that the family of curves f(x,y)=c satisfies the differential equation df = 0.

$$\frac{\delta f}{\delta x}dx + \frac{\delta f}{\delta y}dy = 0$$

So if there exists a function f(x,y) such that:

$M(x,y) = \frac{\delta f}{\delta x}$ and $N(x,y) = \frac{\delta f}{\delta y}$

Then $M(x,y)dx + N(x,y)dy$ is called an exact differential and the equation:

$M(x,y)dx + N(x,y)dy = 0$ is said to be an **exact equation**

EXAMPLE

Solve the differential equation $(1 - 2xy)dx + (4y^3 - x^2)dy = 0$

SOLUTION

With $M(x, y) = 1 - 2xy$ and $N(x, y) = 4y^3 - x^2$, we notice that:

$$\frac{\delta M}{\delta y} = -2x = \frac{\delta N}{\delta x}$$

So the given functions M and N pass the test for exactness. All that is left to do is to find the function f(x,y) such that:

$$\frac{\delta f}{\delta x} = M \text{ and } \frac{\delta f}{\delta y} = N$$

The way to accomplish this is to integrate M with respect to x, integrate N with respect to y, and then merge the results:

$$\int M(x,y)dx = \int (1-2xy)dx = x - x^2y + (\text{a function of y alone})$$

$$\int M(x,y)dx = \int (4y^3 - x^2)dy = y^4 - x^2y + (\text{a function of x alone})$$

These calculations imply that a function f(x,y) which satisfies both:

$$\frac{\delta f}{\delta x} = M = 1 - 2xy \text{ and } \frac{\delta f}{\delta y} = N = 4y^3 - x^2$$

$$f(x,y) = x - x^2y + y^4$$

Therefore, the exact equation we were given is satisfied by the family of curves:

$$x - x^2y + y^4 = c$$

NON-EXACT EQUATIONS AND INTEGRATING FACTORS

Let $M(x,y)dx + N(x,y)dy = 0$ be inexact. We need to ensure that the differential equation is exact. To do this, multiply the equation throughout by an **integrating factor**

$$\mu = \mu(x,y)$$

to obtain:

$$\mu M(x,y)dx + \mu N(x,y)dy = 0 \quad \cdots(\cdot)$$

which should be exact. Then for $(\cdot)$ to be exact we need:

$$\frac{\delta(\mu M)}{\delta y} = \frac{\delta(\mu N)}{\delta x}$$

we obtain by the product rule:

$$\mu\frac{\delta M}{\delta y} + M\frac{\delta \mu}{\delta y} = \mu\frac{\delta N}{\delta x} + N\frac{\delta \mu}{\delta x}$$

Rearranging, we have:

$$M\frac{\delta \mu}{\delta y} - N\frac{\delta \mu}{\delta x} = \mu(\frac{\delta N}{\delta x} - \frac{\delta M}{\delta y}) \quad \cdots(..)(GoldenRule)$$

Simply put:

$$M\mu_y - N\mu_x = \mu(N_x - M_y)$$

To lessen up our work, we want

$$\mu = \mu(x) \text{ or } \mu = \mu(y)$$

If

$$\mu = \mu(x), \frac{\delta \mu}{\delta y} = 0; \text{ or if } \mu = \mu(y), \frac{\delta \mu}{\delta x} = 0$$

Thus $(\cdot\cdot)$ called the **Golden Rule** is much more reduced to solve μ, where two cases arises;

Case 1:

If $\frac{M_y - N_x}{N}$ is a function of x alone, call this function $\xi(x)$.

Then $\mu(x) = e^{\int \xi(x)dx}$ is an integrating factor of $(\cdot)$

Case 2:

If $\frac{M_y - N_x}{-M}$ is a function of y alone, call this function $\psi(x)$.

Then $\mu(y) = e^{\int \psi(x)dx}$ is an integrating factor of $(\cdot)$

NOTE

If a non-exact equation has a solution, then an integrating factor is guaranteed to exist, but that does not mean it is easy to find.

EXAMPLE

Solve the differential equation $\sin 2x + y' \sin^2 x = e^{-y}$

SOLUTION

$$M(x, y) = \sin 2x - e^{-y} \ , \ N(x, y) = \sin^2 x$$

For exactness,

$$\frac{\delta M}{\delta y} = e^{-y} \text{ and } \frac{\delta N}{\delta x} = 2 \sin x \cos x = \sin 2x$$

Clearly, $\frac{\delta M}{\delta y} \neq \frac{\delta N}{\delta x}$

$\therefore$ the differential equation is not exact

Let μ be a function of x alone, that is $\mu = \mu(x)$

Then $\frac{\delta \mu}{\delta y} = 0$ and $M\mu_y - N\mu_x = \mu(N_x - M_y)$ becomes:

$$-\sin^2 x\frac{\delta\mu}{\delta x} = \mu(\sin 2x - e^{-y})$$

Since y still appears, we discard the assumption of $\mu = \mu(x)$

The next option is $\mu = \mu(y)$

Then $\frac{\delta\mu}{\delta y} = 0$ and $M\mu_y - N\mu_x = \mu(N_x - M_y)$ becomes:

$$(\sin 2x - e^{-y})\frac{\delta\mu}{\delta y} = \mu(\sin 2x - e^{-y})$$

$$\Rightarrow \frac{\delta\mu}{\delta y} = \mu$$

which separating to give:

$\frac{\delta\mu}{\mu} = \delta y$ or $\ln\mu = y$ or $\mu = e^y$ we obtain the exact differential equation:

$$e^y \sin 2x - 1 + y' e^y \sin^2 x = 0$$

which is equivalent to : $M(x,y) + N(x,y)y' = 0$

$$M(x,y) = e^y \sin 2x - 1 \ , \ N(x,y) = e^y \sin^2 x$$

$$\frac{\delta f}{\delta x} = M(x,y) = e^y \sin 2x - 1 \ , \ \frac{\delta f}{\delta y} = e^y \sin^2 x$$

$$f(x,y) = \int e^y \sin^2 x = e^y \sin^2 x + \phi(x)$$

However, $2e^y \sin x \cos x + \phi^{'}(x) = e^y \sin 2x - 1$

$$2e^y \sin 2x + \phi^{'}(x) = e^y \sin 2x - 1$$

$$\Rightarrow \phi^{'}(x) = -1 \Rightarrow \phi(x) = -x + k$$

Then $f(x,y) = e^y \sin 2x - x + k = c$

Finally $e^y \sin 2x - x = A$, $A = c - k$; arbitrary constant.

EXERCISE

Solve $x\frac{dy}{dx} - y = 0$

Use exactness and verify directly with separation of variables

LINEAR EQUATIONS

Another type of most occurring differential is the linear type. The form of the linear differential equation as follows:

$$A(x)y' + B(x)y = C(x) \cdots (*)$$

$$A(x) \neq 0$$

Then we can put (*) in the standard form to obtain:

$$y' + p(x)y = q(x) \cdots (**)$$

$$p(x) = \frac{B(x)}{A(x)} \text{ and } q(x) = \frac{C(x)}{A(x)}$$

We recognize (**) as an inexact differential equation which requires an integrating factor $\mu(x)$, that is:

$$\frac{d\mu(x)}{dy} = 0$$

Multiplying through (**) by $\mu(x)$ to obtain the following:

$$\mu(x)y'(x) + \mu[p(x)y(x) - q(x)] = 0 \cdots (***)$$

$$M(x,y) = \mu(x)[p(x)y - q(x)] \text{ and } N(x,y) = \mu(x)$$

If (***) is exact, then $\frac{\delta M}{\delta y} = \frac{\delta N}{\delta x}$ Thus,

$$\frac{\delta}{\delta y}[\mu py - \mu q] = \frac{\delta}{\delta x}\mu(x) = \frac{d\mu(x)}{dx}$$

$$\Rightarrow \mu p = \frac{d\mu}{dx}$$

Separating variables:

$$\int \frac{d\mu}{\mu} = \int p dx \Rightarrow \ln \mu = \int p dx$$

$$\mu = e^{\int p dx}$$

We have the exact equation:

$$\mu y' + \mu p y = \mu q$$

$$\mu y' + \mu' y = \mu q \ (**) \text{ that is } \mu p = \mu'$$

$$\frac{d}{dx}(\mu p) = \mu p$$

Integrating both sides:

$$\mu y = \int \mu q dx + c$$

$$\Rightarrow y = \mu^{-1} \int \mu q dx + \mu^{-1} c$$

$$\Rightarrow y = e^{-\int p(x)dx} \int e^{\int p(x)dx} q(x)dx + ce^{-\int p(x)dx}$$

The solution is in 2 parts in the R.H.S. The term $\mu^{-1}c$, called the **Complementary function** is the solution of the reduced equation:

$$y' + p(x)y = 0$$

obtained by setting $q(x) = 0$ in $(**)$. The differential equation which has $q(x) = 0$ is called **Homogenous differential equation**

The other term on the R.H.S of the solution is dependent on q(x), that $\mu^{-1}\int \mu q dx$ is called **the particular integral** and it satisfies the differential equation:

$$y' + p(x)y = q(x)$$

called the **Non-Homogeneous differential equation**

Notice that the particular integral contains no constant

Let the complementary function be r and the particular integral be s.

Then from the theory above:

$$r' + pr = 0$$

$$s' + ps = q$$

adding:

$$(r' + s') + p(r + s) = q$$

Hence $y = r + s$ is the most general solution of the linear differential equation.

EXAMPLE

Find the solution of $\frac{dy}{dx} = 5x - \frac{3y}{x}$ such that $y(1) = 2$

SOLUTION

First, we rearrange the equation to put it in the standard form of a linear equation:

$$\frac{dy}{dx} + \frac{3}{x}y = 5x$$

Multiply both sides of this by the integrating factor:

$$\mu(x) = e^{\int p(x)dx} = e^{\int \frac{3}{x}} = e^{3logx} = x^3$$

gives the equivalent equation:

$$x^3\frac{dy}{dx} + 3x^2y = 5x^4$$

Notice that the left-hand side is $\frac{d}{dx}(x^3y)$

so by integrating both sides, we get:

$$x^3y = \int (5x^4)dx = x^5 + c$$

$$\Rightarrow y = x^5 + cx^{-3}$$

Now, to satisfy the initial condition, y(1)=2, the value of c must be 1.

Therefore, the solution of the IVP is:

$$y = x^5 + x^{-3}$$

EXERCISE

Solve the IVP : $\frac{dy}{dx} + \frac{3}{x}y = x^2$ given $y(1) = \frac{1}{6}$

EQUATIONS REDUCIBLE TO LINEAR FORM

There are types, which are not linear but can be converted into linear form and solved accordingly,

Consider $(xs(y) + t(y))y' = \mu(y) \cdots (*)$

in which the coefficient of y' is a linear function of x and No other x appears anywhere again.

Thus (*) can be expressed as a linear differential equation with x as the dependent variable and y the independent variable. To be able to do this, multiply (*) through by $\frac{dx}{dy}$ to obtain:

$$\mu(y)\frac{dx}{dy} = xs(y) + t(y) \text{ or } \mu(y)\frac{dx}{dy} - s(y)x = t(y)$$

Thus this is linear differential equation of x as a function of the independent variable y.

EXAMPLE

Solve the differential equation $(3x - 4y^3)y' + y = 0$

SOLUTION

Clearly the coefficient of y' is linear in x and further more x appears nowhere again. Therefore we multiply through by $\frac{dx}{dy}$ to obtain:

$$(3x - 4y^3) + y\frac{dx}{dy} = 0$$

$$\Rightarrow y\frac{dx}{dy} + 3x = 4y^3$$

Whose standard form is:

$$\frac{dx}{dy} + \frac{3}{y}x = 4y^2$$

Integrating factor: $\mu(y) = e^{3\int \frac{dy}{y}} = e^{\ln y^3} = y^3$

$$\Rightarrow y^3\frac{dx}{dy} + 3y^2x = 4y^5$$

$$\int d(xy^3) == \int 4y^5 \Rightarrow xy^3 = \frac{4}{6}y^6 + c$$

$$\Rightarrow 3xy^3 = 2y^6 + c$$

BERNOULLI'S EQUATION

The equation $y' + p(x)y = q(x)y^n \cdots (*)$ differs from a standardized linear differential equation in the R.H.S. Here there is a multiply of y^n , $n \neq 0$, $n \neq 1$

Then divide (*) through by y^n and substitute $v = y^{1-n}$

$$y'y^{-n} + p(x)y^{1-n} = q(x)$$

$$v' = \frac{dv}{dx} == (1-n)y^{-n}y'$$

$$\Rightarrow \frac{v'}{1-n} = y^{-n}y'$$

The equation becomes:

$$v' + (1-n)p(x)v = (1-n)q(x)$$

The solution is obtained using linear equation solution techniques.

EXAMPLE

Solve the differential equation $xy' + y = x^4y^3e^{\frac{1}{2}x^2}$

SOLUTION

Divide through by xy^3

$$y^{-3}y' + x^{-1}y^{-2} = x^3e^{\frac{1}{2}x^2}$$

Let $v = y^{-2}$

$$\Rightarrow \frac{dv}{dx} = -2y^{-3}\frac{dy}{dx}$$

$$\Rightarrow \frac{dv}{-2dx} = y^{-3}\frac{dy}{dx}$$

$$\Rightarrow \frac{dv}{-2dx} + \frac{v}{x} = x^3e^{\frac{1}{2}x^2}$$

$$\Rightarrow \frac{dv}{dx} - \frac{2v}{x} = -2x^3e^{\frac{1}{2}x^2}$$

$$\frac{1}{x^2}\frac{dv}{dx} - \frac{2v}{x^3} = \frac{-2x^3 e^{\frac{1}{2}x^2}}{x^2}$$

$$\Rightarrow \frac{d}{dx}[vx^{-2}] = -2xe^{\frac{1}{2}x^2}$$

$$\Rightarrow \int d(vx^{-2}) = -\int 2xe^{\frac{1}{2}x^2}dx$$

$$\Rightarrow vx^{-2} = -2e^{\frac{1}{2}x^2} + c$$

$$y^{-2}x^{-2} = -2e^{\frac{1}{2}x^2} + c$$

$$1 = y^2x^2(-2e^{\frac{1}{2}x^2} + c)$$

EXERCISE

Solve the following differential equation.

1. $xy' + 2y = xe^x$
2. $(x - y^2 + 2)y' = 1$
3. $y' - 2y = \sqrt{y}\cos x$

RICCATI'S EQUATION

The equation $\frac{dy}{dx} = P(x) + Q(x)y + R(x)y^2 \cdots (*)$ is a non-linear and needs a particular solution before one can have the chance to solve it.

Consider the new function z defined by:

$$z = \frac{1}{y - y_1}$$

where y_1 is the particular solution. Then easy calculations give:

$$\frac{dz}{dx} = -[Q(x) + 2y_1R(x)]z - R(x)$$

Which is linear equation satisfied by new function z. Once it is solved, we go back to the relation $y = y_1 + \frac{1}{z}$

EXAMPLE

Solve the equation $\frac{dy}{dx} = -2 - y + y^2$ where $y_1 = 2$ is a particular solution.

SOLUTION

First we verify that $y_1 = 2$ is a particular solution ,otherwise our calculation is fruitless.

Let

$$z = \frac{1}{y-2} \Rightarrow y = 2 + \frac{1}{z}$$

$$y' = -\frac{z'}{z^2}$$

Hence from the equation satisfied by y, we get :

$$-\frac{z'}{z^2} = -2 - (2 + \frac{1}{z}) + (2 + \frac{1}{z})^2$$

$$\Rightarrow -\frac{z'}{z^2} = \frac{3}{z} + \frac{1}{z^2}$$

Hence

$$z' = -3z - 1$$

This is a linear equation. The general solution is given by:

$$z = e^{-3x}(-\frac{1}{3}e^{3x} + c) = -\frac{1}{3} + ce^{-3x}$$

Therefore, we have:

$$y = 2 + \frac{1}{-\frac{1}{3} + ce^{-3x}}$$

EXERCISE

Verify that $y_1 = \sin x$ is a solution to

$$\frac{dy}{dx} = \frac{2\cos^2 x - \sin^2 x + y^2}{2\cos x}$$

and solve the differential equation using the initial condition $y(0) = -1$

LINEAR DIFFERENTIAL EQUATION OF HIGHER ORDER

INTRODUCTION

A general linear equation of order n is:

$$a_0)(x)\frac{d^n y}{dx^n} + a_1)(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1})(x)\frac{dy}{dx} + a_n(x)y = f(x) \cdots (1)$$

where $a_0(x) \neq 0$, $a_i(x)$, $i = 0, 1, \cdots, n-1, n$; $f(x)$ are given functions of x or are constants.

The equation (1) is homogeneous when f(x)= 0 and non-homogeneous when $f(x) \neq 0$.

The solution of the homogeneous differential equation is called the **Complimentary function** and the solution of the non-homogeneous is known as the **Particular Integral**.

Then the **general solution =complimentary function + particular integral**

HOMOGENEOUS EQUATION WITH CONSTANT COEFFICIENTS

A general homogeneous differential equation with constant coefficients is:

$$a_0\frac{d^n y}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_{n-1}\frac{dy}{dx} + a_n y = 0 \cdots (2)$$

For $n = 2$, we have $a_0\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2 y = 0 \cdots (3)$

Consider at trial solution $y = e^{mx}$, m to be determined for the constant coefficient linear differential equation. Then (3) becomes:

$$(a_0 m^2 + a_1 m + a_2)e^{mx} = 0 \cdots (4)$$

Clearly $e^{mx} \neq 0$, hence $a_0 m^2 + a_1 m + a_2 = 0$

Equation (4) is known as the **Auxiliary polynomial equation**.

The roots of this polynomial determine the solutions of the original differential equation.

There are three cases to consider:

Case 1:

If the roots are real and distinct, that is,

$$m = m_1, m = m_2 \text{ and } m_1 \neq m_2$$

where then the general solution of the differential equation is:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

Case 2:

If the roots are real and identical, that is,

$$m = m_1 = m_2 \text{ if we denote the double root by } m$$

where then the general solution of the differential equation is:

$$y = c_1 e^{mx} + c_2 x e^{mx}$$

In this case, for an nth order linear differential equation with constant coefficients

if $m = m_1 = m_2$ is of multiplicity k

then the root $m = m_1$ (k repeating)

$$y_k = (c_1 + c_2 x + c_3 x^2 + \cdots + c_k x^{k-1}) e^{m_1 x}$$

Case 3:

If the roots are not real. In this case, the roots are complex conjugates. If we denote them by

$$m_1 = \alpha + \beta i \text{ and } m_2 = \alpha - \beta i$$

then the general solution of the differential equation is:

$$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$$

EXAMPLE 1

Solve the differential equation $y'' + 3y' + 2y = 0$

Solution

Let $y = e^{mx}, y' = me^{mx}, y'' = m^2e^{mx}$:

$$\Rightarrow m^2e^{mx} + 3me^{mx} + 2e^{mx} = 0$$

$$\Rightarrow m^2 + 3m + 2 = 0 : \ m = -1, -2$$

$$\Rightarrow y = c_1e^{-x} + c_2e^{-2x}$$

EXAMPLE 2

Solve $\frac{d^2y}{dx^2} + -6\frac{dy}{dx} + 9y = 0$

Solution

$$\text{Aux} : m^2 - 6m + 9 = 0$$

$$\Rightarrow (m-2)^2 = 0$$

$$\Rightarrow m = 3$$

$$\Rightarrow y = c_1 e^{3x} + c_2 x^{3x}$$

EXAMPLE 3

Solve the differential equation $y'' + y' + y = 0$

Solution

$$\text{Aux} : m^2 + m + 1 = 0$$

$$m = -\frac{1}{2} \pm \frac{i}{2}\sqrt{3}$$

$$y = e^{-\frac{1}{2}x}\left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x\right)$$

FUNDAMENTAL SETS OF SOLUTIONS

Consider a homogeneous second order linear differential equation

$$y'' + p(x)y' + q(x)y = 0 \cdots (*)$$

Principle of Superposition

Let y_1 and y_2 be two solutions of the homogeneous linear equation(*) on the interval I. If c_1 and c_2 are constants, then the linear combination

$$y = c_1y_1 + c_2y_2 \cdots (**)$$

is also a solution on I. Prove as exercise.

LINEAR INDEPENDENCE OF TWO FUNCTIONS AND THE WRONSKIAN

Two functions defined on an open interval are called linearly independent provided that neither is a constant multiple of the other. Two functions are said to be linearly dependent if they are not linearly independent; that is, one is a constant multiple of the other.

EXAMPLE

$y_1 = \cos x$ and $y_2 = \sin x$ are two solutions of the equation $y'' + y = 0$. By principle of superposition $y = 3y_1 - 2y_2 = 3\cos x - 2\sin x$ is also a solution.

We can always determine whether two given functions f and g are linearly dependent on an interval I by nothing at a glance whether either of the two quotients $f|g$ and $g|f$ as a constant on I.

The following pairs are linearly independent on the entire real line:

$(\sin x, \cos x), (e^x, e^{-2x})$ and $(x.|x|)$

Given two functions f and g, the **Wronskian** of f and g is the determinant

$$W(f,g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = fg' - f'g$$

For example,

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$$

while

$$W(e^x, xe^x) = \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix} = e^{2x} \neq 0 \quad \forall x$$

Given two functions f(x) and g(x) that are differentiable on some interval I:

1. If $W(f,g)(x) \neq 0$ for some $x_0$in I, then f(x) and g(x) are linearly independent on the interval F
2. If f(x) and g(x) are linearly dependent on I then W(f,g)(x) = 0 for all x is the interval.

NOTE

It is possible for two linearly independent functions to have a zero Wronskian.

Suppose that y_1 and y_2 are two solutions of homogeneous second order linear equation $y'' + pxy' + q(x)y = 0$ on open interval I on which p(x) and q(x) are continuous. If $W(y_1, y_2) \neq 0$, then the two solutions are called a **fundamental set of solutions** and the general solution is

$y = c_1 y_1 + c_2 y_2$

REDUCTION OF ORDER

Consider the non-constant coefficient, second order differential equations of the form

$$y'' + pxy' + q(x)y = 0 \cdots (a)$$

In general, finding solutions to these kinds of differential equations can be much more difficult than finding solutions to constant coefficient differential equation.

Solution of Reduction Order:

Consider

$$y'' + pxy' + q(x)y = 0$$

Let

$$y_2(x) = v(x)y_1(x) \cdots (b)$$

We begin by substituting the expression in (a) and (b), using the derivatives:

$$y_2^{'} = vy_1^{'} + v'y_1 \Rightarrow y_2^{''} = vy_1^{''} + 2v'y_1^{'} + v''y_1$$

We get:

$$[vy_1^{''} + 2v'y_1^{'} + v''y_1] + p[vy_1^{'} + v'y_1] + qvy_1 = 0$$

but $y'' + pxy' + q(x)y = 0$ if y_1 is a solution of (a).
This leaves the equation $y_1v'' + (2y_1^{'} + py_1)v' = 0 \cdots (c)$
The key to success of this method is that (c) is linear in v'. Thus the substitution of (b) has reduced the second order linear equation in (a) to the first order in v' in linear equation in (c).

If we write u = v' and assume that y_1 never vanishes on I, then (c) yields:

$$u' + (\frac{2y_1^{'}}{y_1} + p(x))u = 0 \cdots (d)$$

An integrating factor for (c) is:

$$\mu = \exp\{\int [2\frac{y_1^{'}}{y_1} + p(x)]dx\} = \exp[2\ln|y_1| + \int p(x)dx]$$

Thus

$$\mu = y_1^2 e^{\int p(x)dx}$$

We now integrate the equation in (d) to obtain:

$$uy_1^2 e^{\int p(x)dx} = c$$

So

$$v' = u = \frac{c}{y_1^2} e^{-\int p(x)dx}$$

Another integration now gives:

$$\frac{y_2}{y_1} = v = c \int \frac{e^{\int p(x)dx}}{y_1^2} dx + k$$

With the particular choices c=1 and k=0, we get

$$y_2 = y_1 \int \frac{e^{\int p(x)dx}}{y_1^2} dx$$

EXAMPLE

Find the general solution to $2x^2y'' + xy' - 3y = 0$ given that $y_1(x) = x^{-1}$ is a solution.

Solution

$$2x^2y'' + xy' - 3y = 0$$

$$\Rightarrow y'' + \frac{1}{2x}y' + \frac{3}{2x^2}y = o \text{ where } p(x) = \frac{1}{2x}$$

$$y_2 = y_1 \int \frac{e^{\int p(x)dx}}{y_1^2}dx$$

$$\Rightarrow y_2 = x^{-1} \int \frac{e^{\int \frac{1}{2}dx}}{(x^{-1})^2} = x^{-1} \int \frac{x^{-\frac{1}{2}}}{x^{-2}}dx$$

$$\Rightarrow y_2 = x^{-1}\int x^{\frac{3}{2}}dx = x^{-1}(\frac{2}{5}x^{\frac{5}{2}} + c)$$

Let c= 0

$$y_2 = \frac{2}{5}x^{\frac{3}{2}}$$

Thus a general solution is

$$y = c_1x^{-1} + c_2x^{\frac{3}{2}}$$

LINEAR NON-HOMOGENEOUS EQUATION WITH CONSTANT COEFFICIENTS

A general linear non-homogeneous constant coefficient equation is:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = f(x) \cdots (e)$$

f(x) is a given function of x and $a_0, a_1, \cdots, a_{n-1}, a_n$ also given constants.

We study three methods to find our particular integral ; which are mostly

1. Method of Undetermined Coefficients
2. Method of Variation of Parameter
3. Method of the differential Operators

METHOD OF UNDETERMINED COEFFICIENTS

A trial solution function $y_p(x)$ is assumed.Its form depends on f(x) of (e) $y_p(x)$ contains a finite number of arbitrary constants.$y_p(x)$ is then substituted into (e) and the constants takes values which satisfies (e) completely.

For $f(x) = ae^{bx}$ a,b are constants, if the auxiliary equation of (e) has m=b as a root of multiplicity k, then the trial function is of the form $y_p(x) = Ax^k e^{bx}$, where A is the constant to be determined. If m=b is not a root, take k=0.

Example

Find a particular integral of $y'' - 3y' + 2y = e^{2x}$

Solution

The auxiliary equation is: $m^2 - 3m + 2 = 0$

$$(m-2)(m-1) = 0 \Rightarrow m = 2, 1$$

Hence $f(x) = e^{2x}$ that is b=2, a root.

The trail solution is $y_p(x) = Axe^{2x}$

Substituting this into the differential equation

$$y_p(x) = Axe^{2x}, y_p'(x) = Ae^{2x} + 2Axe^{2x}, y_p''(x) = 2Ae^{2x} + 2Ae^{2x} + 4Ae^{2x}$$

$$\Rightarrow 4Ae^{2x} + 4Ae^{2x} - 3Ae^{2x} - 6Axe^{2x} + 2Axe^{2x} = e^{2x}$$

$$\Rightarrow Ae^{2x} = e^{2x} \Rightarrow A = 1$$

$$\therefore y_p(x) = xe^{2x}$$

For $f(x) = a\sin bx$ or $\cos bx$ or both, a,b constants, if $m^2 + b^2$ is a factor of the auxiliary equation of multiplicity k, then:

$$y_p(x) = x^k(A\sin bx + B\cos bx)$$

where A,B are constant to be determined.
If $m^2 + b^2$ is not a factor, then k=0

Example

Solve the differential equation $y'' + 4y = 3\sin 2x$

Solution

$$m^2 + 2^2 = 0,\ b = 2, k = 1$$

$$y_p = A\sin 2x + Bx\cos 2x \cdots (i)$$

$$y_p^{'} = A(\sin 2x + 2x\cos 2x) + B(\cos 2x - 2x\sin 2x) \cdots (ii)$$

Substitute back into the equation gives:

$\Rightarrow A[2\cos 2x + 2(\cos 2x - 2x\sin 2x)] + B[-2\sin 2x - 2(\sin 2x + 2x\cos 2x)] + 4[A\sin 2x + Bx\cos 2x] = 3\sin 2x$

$$\Rightarrow 4A\cos 2x - 4B\sin 2x = 3\sin 2x$$

Comparing the coefficients:

$$A = 0, B = -\frac{3}{4}$$

$$y_p(x) = -\frac{3}{4}x\cos 2x$$

General solution: $y(x) = c_1\cos 2x + c_2\sin 2x - \frac{3}{4}x\cos 2x$

For $f(x) = ax^s$,a,s(integer)constant,if the auxiliary equation has m=0 as a root of multiplicity k,then trial function:

$$y_p(x) = x^k(Ax^s + Bx^s + \cdots + Px + Q)$$

where $A, B, \cdots, P, Q$ are constants to be determined.

If m=0 is not a root, then k=0

Example Find a particular solution of $y'' + 4y = 3x^3$

Solution

$y_p = Ax^3 + Bx^2 + Cx + D$

$y_p^{'} = 3Ax^2 + 2Bx + C$

$y_p^{''} = 6Ax + 2B$

Substituting in the differential equation, we get:

$$\Rightarrow (6Ax + 2B) + 4(Ax^3 + Bx^2 + Cx + D) = 3x^3$$

Solving for the various terms we get:

$$A = \frac{3}{4}, B = 0, C = -\frac{9}{8} \text{ and } D = 0$$

Hence the particular solution of the differential equation is

$$y_p = \frac{3}{4}x^3 - \frac{9}{8}x$$

If f(x) is the sum of any two or all these special functions, the particular integral is the appropriate sum of individual particular integral.

Hence f(x) takes the trial solution:

$$y_p = x^k[(A_0+A_1(x)+\cdots+A_nx^n)e^{mx}\cos bx+(B_0+B_1(x)+\cdots+B_nx^n)e^{mx}\sin bx]$$

Example Determine the appropriate form for a particular solution of $y'' + 6y' + 13y = e^{-3x}\cos 2x$

Solution

The characteristic equation $m^2 + 6m + 13 = 0$ has roots:

$$m = -3 \pm 2i$$

So the complementary function is:

$$y_c = e^{-3x}(A\cos 2x + B\sin 2x)$$

This is the same form as a first attempt $e^{-3x}(A\cos 2x + B\sin 2x)$ at a particular solution, so we must multiply by x to eliminate duplication. Hence we would take $y_p = e^{-3x}(Ax\cos 2x + Bx\sin 2x)$

METHOD OF VARIATION OF PARAMETERS

Consider a non homogeneous linear second order differential equation of the form:

$$y'' + p(x)y' + q(x)y = f(x) \cdots (I)$$

Suppose y_1 and y_2 are two solutions of the homogeneous equation

$$y'' + p(x)y' + q(x)y = 0 \cdots (II)$$

Then the complimentary function is:

$$y(x) = c_1 y_1(x) + c_2 y_2(x) \cdots (III)$$

where c_1 and c_2 are arbitrary constants

To find the particular integral, this method requires that the trial function

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x) \cdots (IV)$$

where we replace the constants or parameters c_1 and c_2 in the complimentary function in (III) by variables: functions u_1 and u_2 of x.
(IV) is substituted into (I) to determine $u_1(x)$ and $u_2(x)$.
To determine $u_1(x)$ and $u_2(x)$ the following conditions should be fulfilled:

1. (IV) must satisfy (I)
2. A condition imposed arbitrary to facilitate the calculation of the function $u_1(x)$ and $u_2(x)$

$$y_p(x) = u_1y_1 + u_2y_2 \cdots (IV*)$$

$$y_p^{'}(x) = u_1^{'}y_1 + u_1y_1^{'} + u_2^{'}y_2 + u_2y_2^{'} \cdots (V)$$

Let $u_1^{'}y_1 + u_2^{'}y_2 = 0 \cdots (VI)$: **Condition imposed**

$$\Rightarrow y_p^{'}(x) = u_1 y_1^{'} + u_2 y_2^{'}$$

$$\Rightarrow y_p^{''}(x) = u_1^{'} y_1^{'} + u_1 y_1^{''} + u_2^{'} y_2^{'} + u_2 y_2^{''}$$

Substituting (VI),(V) and (IV*) into (I). We have

$\Rightarrow u_1^{'} y_1^{'} + u_1 y_1^{''} + u_2^{'} y_2^{'} + u_2 y_2^{''} + p(u_1 y_1^{'} + u_2 y_2^{'}) + q(u_1 y_1 + u_2 y_2) = f(x)$

$\Rightarrow u_1(y_1^{''} + p y_1^{'} + q y_1) + u_2(y_2^{''} + p y_2^{'} + q y_2) + u_1^{'} y_1^{'} + u_2 y_2^{'} = f(x)$

But both y_1 and y_2 satisfy the homogeneous equation (II) associated with the non-homogeneous (I)

So $y_1^{''} + p y_1^{'} + q y_1 = 0$ and $y_2^{''} + p y_2^{'} + q y_2 = 0$

Then solve the following two equations for u_1 and u_2:

$$u_1^{'} y_1 + u_2^{'} y_2^{'} = 0$$

$$u_1^{'} y_1^{'} + u_2^{'} y_2^{'} = f(x)$$

The Cramer's rule for the solution of a linear system of equation yields:

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} \qquad u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

$$\Rightarrow u_1' = \frac{-y_2 f(x)}{W(y_1, y_2)}, \qquad \Rightarrow u_2' = \frac{y_1 f(x)}{W(y_1, y_2)}$$

since the denominator is the Wronskian.

$$\Rightarrow u_1 = \int \frac{y_2 f(x)}{W(y_1, y_2)} dx, \qquad \Rightarrow u_2 = \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

$$y_p = -y_1(x) \int \frac{y_2 f(x)}{W(y_1, y_2)} dx + y_2(x) \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

DIFFERENTIAL OPERATOR METHOD(D-OPERATOR)

This method is very efficient for the solution of a general linear equation of nth order and with constant coefficients. It is most efficient because it can be manipulated algebraically;

If

$$D = \frac{d}{dx} \qquad D^2 = \frac{d^2}{dx^2}, \text{ etc}$$

$$D(x^2) = 2x \quad D^2(e^{3x}) = 9e^{3x}$$

The differential operator of order n is in the form:

$$f(D)y = a_0 D^n y + a_1 D^{n-1} y + \cdots + a_{n-1} Dy + a_n y$$

FUNDAMENTAL PROPERTIES OF THE D-OPERATOR

The following properties are satisfied by the D-operator.

1. $D^n(u+v) = D^n u + D^n v$ (distributive property)

FACTORABILITY

Any constant coefficient nth order linear differential equation can be written as : $f(D)[y] = R(x)$

where $f(D)$is a polynomial function of the operator D

Example: $y'' - 3y' + 2y = D^2y - 3Dy + 2y = e^{2x}$

$$\Rightarrow (D^2 - 3D + 2)y = e^{2x}$$

$$\Rightarrow f(D) = D^2 - 3D + 2$$

LINEAR EQUATIONS OF 1st - ORDER

Recall the familiar 1st-Order linear equation with constant coefficient

$$\frac{dy}{dx} - ay = R(x), \text{ a is a constant.}$$

which is equivalent to: $(D - a)y = R(x)$

$$\Rightarrow y = \frac{1}{D - a}R(x) \cdots (*)$$

The integrating factor $I = e^{-\int a dx} = e^{-ax}$

multiplying both sides by the integrating factor we obtain

$$e^{-ax}\frac{dy}{dx} - ae^{-ax}y = e^{-ax}R(x)$$

$$\Rightarrow \frac{d}{dx}(ye^{-ax}) = e^{-ax}R(x)$$

$$\Rightarrow (ye^{-ax}) = \int e^{-ax} R(x)$$

$$\Rightarrow y = e^{-ax} \int e^{-ax} R(x)$$

From *

$$\Rightarrow \frac{1}{D-a} R(x) = e^{-ax} \int e^{-ax} R(x)$$

Therefore if we have $(D-2)y = e^{2x} \Rightarrow y = \frac{1}{D-2} e^{2x}$

Then $y = e^{2x} \int e^{-2x} \cdot e^{2x} dx$

$$\Rightarrow y = e^{2x} \int dx = e^{2x}(x+c) = xe^{2x} + ce^{2x}$$

LINEAR EQUATION OF ORDER OTHER THAN ONE

Here we discuss second order linear equations.

$$(aD^2 + bD + c)y = f(x)$$

Example: Solve $(D^2 - 3D + 2)y = e^{2x}$

Solution

$$(D^2 - 3D + 2)y = e^{2x} = (D - 2)(D - 1)y = e^{2x}$$

Let

$$(D - 1)y = z$$

then

$$(D - 2)z = e^{2x}$$

$$z = \frac{1}{D - 2}e^{2x} = e^{2x}\int e^{-2x} \cdot e^{2x}dx = e^{2x}(x + c_1)$$

Thus $(D-1)y = e^{2x}(x+c_1)$

$$\therefore y = \frac{1}{D-1}[e^{2x}(x+c_1)] = e^{2x}\int e^{-x}[e^{2x}(x+c_1)]dx$$

$y = xe^{2x} + Ae^{2x} + Be^{-x}$, A and B are arbitrary constant.

EXERCISE

1. $D(D-1)y = 5$
2. $(D^2+4)y = 0$
3. $(D^2+D+1)y = 2$
4. $(D^2+9)y = \sin 2x$

LAPLACE TRANSFORMS

The Laplace Transform of a function $F(t)$ is denoted by $\mathcal{L}\{F(t)\}$ and is defined as the integral of $F(t)e^{-st}$ between the limits $t = 0$ and $t = \infty$.

$$\mathcal{L}\{F(t)\} = \int_0^\infty F(t)e^{-st}dt$$

The constant parameter s is assumed to be positive and large enough to ensure that the product $F(t)e^{-st}$ converges to zero as $t \longrightarrow \infty$, for most common functions $F(t)$

In determining the transform of any function, you will appreciate that the limits are substituted for t, so that the result will be a function of s.

$$\mathcal{L}\{F(t)\} = \int_0^\infty F(t)e^{-st}dt = f(s)$$

EXAMPLE 1

Find the Laplace transform of $F(t) = a(constant)$

$$\mathcal{L}\{a\} = \int_0^\infty ae^{-st}$$

$$= a[\frac{e^{-st}}{-s}]_0^\infty$$

$$-\frac{a}{s}[e^{-st}]_0^\infty$$

$$-\frac{a}{s}\{0-1\} = \frac{a}{s}$$

$$\therefore \mathcal{L}\{a\} = \frac{a}{s}$$

EXAMPLE 2

To find the Laplace transform of $F(t) = e^{at}$

$$\mathcal{L}\{e^{at}\} = \int_0^{\infty} e^{at} e^{-st} dt$$

$$= \int_0^{\infty} e^{-} a - stdt$$

$$[\frac{e^{-(a-s)t}}{-(s-a)}]_0^{\infty} = -\frac{1}{s-a}\{0-1\} = \frac{1}{s-a}$$

NOTE

$\mathcal{L}\{a\} = \frac{a}{s}$ $\quad$ $\mathcal{L}\{4\} = \frac{4}{s}$

$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ $\quad$ $\mathcal{L}\{e^{4t}\} = \frac{1}{s-4}$ $\quad$ $\mathcal{L}\{e^{-2t}\} = \frac{1}{s+2}$

EXAMPLE 3

Laplace transform of $F(t) = \sin at$

$\mathcal{L}\{\sin at\} = \int_0^\infty \sin at \cdot e^{-st} dt$

Using integration by parts, however, its much easier to use the fact that $e^{i\theta} = \cos\theta + i \sin\theta$ so that $\sin\theta$ is the imaginary part of $e^{i\theta}$, written $Im(e^{i\theta})$

$$\mathcal{L}\{\sin at\} = \mathcal{L}\{Im(e^{iat})\}$$

$$= Im \int_0^\infty e^{iat} e^{-st} dt$$

$$= Im(\int_0^\infty e^{-(s-ia)t} dt)$$

$$= Im([\frac{e^{-(s-ia)t}}{-(s-ia)}]_0^\infty)$$

$$= Im(-\frac{1}{s-ia}[0-1])$$

$$= Im(\frac{1}{s-ia})$$

$$= Im(\frac{s+ia}{s^2+a^2}) = \frac{a}{s^2+a^2}$$

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$$

Similarly,

$$\mathcal{L}\{\cos at\} = Re(\frac{s+ia}{s^2+a^2})$$

$$= \frac{a}{s^2+a^2}$$

EXAMPLE 4

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4}$$

$$\mathcal{L}\{\cos 4t\} = \frac{s}{s^2 + 16}$$

EXAMPLE 5

To find the transform of $F(t) = t^n$ where n is a positive integer.

By definition $\mathcal{L}\{t^n\} = \int_0^\infty t^n e^{-st} dt$

Integrating by parts:

$$\mathcal{L}\{t^n\} = [t^n(\frac{e^{-st}}{-s})]_0^\infty + \frac{n}{s}\int_0^\infty e^{-st} t^{n-1} dt$$

$$= -\frac{1}{s}[t^n e^{-st}]_0^\infty + \frac{n}{s}\int_0^\infty e^{-st} t^{n-1} dt$$

In a product such as $t^n e^{-st}$, the numerical value of s is large enough to make the product converge to zero as $t \longleftarrow \infty$.

$$\therefore [e^{-st} t^n]_0^\infty = 0 - 0dt$$

$\mathcal{L}\{t^n\} = \frac{n}{s} \int_0^\infty e^{-st} t^{n-1} dt$
Notice that $\int_0^\infty e^{-st} t^{n-1} dt$, is identical to $\int_0^\infty e^{-st} t^n dt$ except that n is replaced by $(n-1)$.

If $I_n = \int_0^\infty e^{-st} t^n dt$, then $I_{n-1} = \int_0^\infty e^{-st} t^{n-1} dt$ and the result becomes $I_n = \frac{n}{s} I_{n-1}$. Replace n by $n-1$ repeatedly.

$$I_n = \int_0^\infty e^{-st} t^n dt$$

$$= \frac{n}{s} I_{n-1} = \frac{n}{s} \cdot \frac{n-1}{s} I_{n-2}$$

$$= \frac{n}{s} \cdot \frac{n-1}{s} I_{n-2} \cdot \frac{n-2}{s} I_{n-3} \qquad etc \cdots$$

$$I_n = \frac{n}{s} \cdot \frac{n-1}{s} I_{n-2} \cdot \frac{n-2}{s} \cdots \frac{n-(n-1)}{s} I_0$$

$$I_0 = \mathcal{L}\{t^0\} = \mathcal{L}\{1\} = \frac{1}{s}$$

$$\therefore I_n = \frac{n(n-1)(n-2)(n-3)\cdots(3)(2)(1)}{s^{n+1}} = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

EXAMPLE 6

1. $\mathcal{L}\{t\} = \frac{1}{s^2}$
2. $\mathcal{L}\{t^2\} = \frac{2}{s^3}$
3. $\mathcal{L}\{t^3\} = \frac{6}{s^4}$

LAPLACE TRANSFORMS OF HYPERBOLIC FUNCTIONS

Laplace transform of $F(t) = \sinh at$ and $F(t) = \cosh at$

$$\sinh at = \frac{1}{2}(e^{at} - e^{-at})$$

$$\cosh at = \frac{1}{2}(e^{at} + e^{-at})$$

$F(t) = \sinh at$

$$\mathcal{L}\{\sinh at\} = \int_0^\infty \sinh at \cdot e^{-st} dt$$

$$= \frac{1}{2}\int_0^\infty (e^{at} + e^{-st}) dt$$

$$= \frac{1}{2}\left[\frac{e^{-(s-a)t}}{-(s-a)} - \frac{e^{-(s+a)t}}{-(s+a)}\right]_0^{\infty}$$

$$= \frac{1}{2}\left[\frac{1}{s-a} - \frac{1}{s+a}\right] = \frac{a}{s^2-a^2}$$

$$\mathcal{L}\{\sinh at\} = \frac{a}{s^2-a^2}$$

Similarly,

$$\mathcal{L}\{\cosh at\} = \frac{s}{s^2-a^2}$$

EXAMPLE 7

1. $\mathcal{L}\{2\sin 3t + \cos 3t\}$

$$= \mathcal{L}\{\sin t\} + \mathcal{L}\{\cos 3t\}$$

$$= 2 \cdot \frac{3}{s^2+9} + \frac{s}{s^2+9}$$

$$= \frac{s+6}{s^2+9}$$

2. $\mathcal{L}\{4e^{2t} + 3\cosh 4t\}$

$$= 4\mathcal{L}\{e^{2t}\} + 3\mathcal{L}\{\cosh 4t\}$$

$$= 4 \cdot \frac{1}{s-2} + 3 \cdot \frac{s}{s^2-16}$$

$$= \frac{4}{s-2} + \frac{3s}{s^2-16} = \frac{7s^2-6s-64}{(s-2)(s^2-16)}$$

EXERCISE

1. $\mathcal{L}\{2\sin 3t + 4\sinh 3t\}$
2. $\mathcal{L}\{2e^{4t} + \cosh 2t\}$
3. $\mathcal{L}\{t^3 + t^2 - 4t + 1\}$

THEOREM 1: THE FIRST SHIFT THEOREM

If $f(t)$ has a transformation $F(s)$ (where $s > k$), then $e^{-at}f(t)$ has the transform $F(s+a)$(where $s+a > k$).

$$\mathcal{L}\{e^{-at}F(t)\} = f(s+a)$$

EXAMPLE 8

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}$$

then

$$\mathcal{L}\{e^{-3t}\sin 2t\} = \frac{2}{(s+3)^2+4} = \frac{2}{s^2+6s+13}$$

Similarly,

$$\mathcal{L}\{t^2\} = \frac{2}{s^3}$$

then

$$\mathcal{L}\{t^2e^{4t}\} = \frac{2}{(s-4)^3}$$

EXERCISE

Find the following :

1. $\mathcal{L}\{e^{-2t}\cosh 3t\}$
2. $\mathcal{L}\{2e^{3t}\sin 3t\}$
3. $\mathcal{L}\{4e^{-t}\}$
4. $\mathcal{L}\{e^{2t}\cos t\}$
5. $\mathcal{L}\{e^{3t}\sinh 2t\}$
6. $\mathcal{L}\{t^3e^{-2t}\}$

THEOREM 2: MULTIPLYING BY t and t^n

If

$$\mathcal{L}\{F(t)\} = f(s)$$

then,

$$\mathcal{L}\{tF(t)\} = -\frac{d}{ds}\{f(s)\}$$

Hence,

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}[F(s)] = (-1)^n F^{(n)}(s)$$

EXAMPLE 9

$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}$

$$\mathcal{L}\{t \sin 2t\} = -\frac{d}{ds}[\frac{2}{s^2+4}]$$

$$= -\frac{(s^2+4)(0) - 2(2s)}{(s^2+4)^2}$$

$$= \frac{4s}{(s^2+4)^2}$$

Similarly, $\mathcal{L}\{t \cosh 3t\}$

$$= -\frac{d}{ds}[\frac{s}{s^2-9}]$$

$$= -\frac{(s^2-9) - s(2s)}{(s^2-9)^2} = \frac{s^2+9}{(s^2-9)^2}$$

Also,

$$\mathcal{L}\{t^2 \cosh 3t\} = \mathcal{L}\{t(t \cosh 3t)\}$$

$$= -\frac{d}{ds}\left(\frac{s^2+9}{(s^2-9)^2}\right)$$

$$= \frac{2s(s^2+27)}{(s^2-9)^2}$$

THEOREM 3: DIVIDING BY t

If $\mathcal{L}\{F(t)\} = f(s)$ and if $\lim_{t \longrightarrow 0, t>0} \frac{f(t)}{t}$ exists, then

$$\mathcal{L}\{\frac{F(t)}{t}\} = \int_0^\infty f(s)ds$$

This rule is somewhat restricted in use, since it is applicable **if and only if** the limit of $[\frac{F(t)}{t}]$ as $t \longrightarrow 0$ exists.

Indeterminate case, we use L'hopital rule to find out.

EXAMPLE 10

Determine $\mathcal{L}\{\frac{\sin at}{t}\}$

SOLUTION

First we test $\lim_{t \longrightarrow 0}[\frac{\sin at}{t}] = \frac{0}{0} = ?$

$$\lim_{t \longrightarrow 0}[\frac{\sin at}{t}] = \lim_{t \longrightarrow 0}[\frac{a\cos at}{1}] = a$$

(i.e the limits exists)

EXAMPLE 11

Determine $\mathcal{L}\{\frac{1-\cos 2t}{t}\}$

SOLUTION

First we text whether $\lim_{t\longrightarrow 0}[\frac{1-\cos 2t}{t}]$ exists

$$\lim_{t\longrightarrow 0}[\frac{1-\cos 2t}{t}] = \frac{1-1}{0} = \frac{0}{0}$$

By L'Hopital's rule

$$\lim_{t\longrightarrow 0}[\frac{1-\cos 2t}{t}] = \lim_{t\longrightarrow 0}[\frac{2\sin 2t}{1}] = 0$$

(i.e the limit exists)

$\mathcal{L}\{1-\cos 2t\} = \frac{1}{s} - \frac{s}{s^2+4}$

By theorem 3,

$\mathcal{L}\{\frac{1-\cos 2t}{t}\} = \int_0^\infty [\frac{1}{s} - \frac{s}{s^2+4}]ds$

$= [\ln s - \frac{1}{2}\ln(s^2+4)]_s^\infty$

$= \frac{1}{2}[\ln(\frac{s^2}{s^2+4})]_s^\infty$

$$s \longrightarrow \infty \qquad \ln\left[\frac{s^2}{s^2+4}\right] \longrightarrow \ln(1) = 0$$

$$\mathcal{L}\left\{\frac{1-\cos 2t}{t}\right\} = -\frac{1}{2}\ln\left[\frac{s^2}{s^2+4}\right]$$

$$\ln\left[\frac{s^2}{s^2+4}\right]^{-\frac{1}{2}} = \ln\sqrt{\frac{s^2+4}{s^2}}$$

EXERCISE

Determine the Laplace Transforms of the following functions

1. $\frac{e^{3t}-1}{t}$
2. $\cos 2t$
3. $e^{3t}\cos 4t$
4. e^{4t}
5. $t^2 - 3t + 4$
6. $6t^2$
7. $t^2 \sin t$
8. $\sinh 3t$

THEOREM 4

Let f be a piecewise-continuous function on $[0, \infty]$ exponential order. A function g is defined by

where a is a positive number, then

$$\mathcal{L}[g] = e^{-as}\mathcal{L}[f(t)]$$

EXAMPLE

Let

$$g(t) = \begin{cases} 0 & 0 \leq t \leq a \\ 1 & a < t \end{cases}$$

Then

$$\mathcal{L}[g] = \frac{1}{s} e^{-as}$$

THEOREM 5

If $f(t)$ is periodic with period T, that is $f(t+T) = f(t)$, then

$$\mathcal{L}[f(t)] = \frac{\int_0^\infty e^{-st} f(t)dt}{1 - e^{-Ts}}$$

INVERSE LAPLACE

Suppose the Laplace transform $\mathcal{L}[F(t)] = f(s)$, then the inverse transform is given as

$$\mathcal{L}^{-1}[f(s)] = F(t).$$

Linearity of inverse Laplace transforms:

$$\mathcal{L}^{-1}[aF(s) + bG(s)] = a\mathcal{L}^{-1}[F(s)] + b\mathcal{L}^{-1}[G(s)]$$

for any constant a, b

EXAMPLES

1. $\mathcal{L}^{-1}[\frac{1}{s-2}] = e^{2t}$
2. $\mathcal{L}^{-1}[\frac{4}{s}] = 4$
3. $\mathcal{L}^{-1}[\frac{s}{s^2+25}] = \cos 5t$
4. $\mathcal{L}^{-1}[\frac{12}{s^2-9}] = 4\sinh 3t$

RULES OF PARTIAL FRACTIONS

1. The numerator must be lower degree than the denominator. If it is not, then we first divide out.
2. Factorize the denominator into prime factors. These determine the shapes of the partial fractions.
3. A linear factor $(s+a)$ gives a partial fraction $\frac{A}{s+a}$ where A is a constant to be determined.
4. A repeated factor $(s+a)^2$ gives $\frac{A}{s+a}+\frac{B}{(s+a)^2}$
5. Similarly $(s+a)^3$ gives $\frac{A}{s+a}+\frac{B}{(s+a)^2}+\frac{C}{(s+a)^3}$
6. A quadratic factor (s^2+ps+q) gives $\frac{Ps+Q}{s^2+ps+q}$
7. Repeated quadratic factors $(s^2+ps+q)^2$ gives $\frac{Ps+Q}{s^2+ps+q}+\frac{Rs+T}{(s^2+ps+q)^2}$

EXAMPLE 13

$\mathcal{L}^{-1}[\frac{5s+1}{s^2-s-12}]$

SOLUTION

$$\frac{5s+1}{s^2-s-12} \equiv \frac{5s+1}{(s-4)(s+3)} \equiv \frac{A}{s-4} + \frac{B}{s+3}$$

$$\frac{5s+1}{s^2-s-12} \equiv \frac{A}{s-4} + \frac{B}{s+3}$$

$$5s+1 \equiv A(s+3) + B(s-4)$$

Let $s = 4$

$$21 = A(7) + B(0) \Rightarrow A = 3$$

Let $s = -3$

$$-14 = A(0) + B(-7) \Rightarrow B = 2$$

$$\therefore \frac{5s+1}{s^2 - s - 12} = \frac{3}{s-4} + \frac{2}{s+3}$$

$$\mathcal{L}^{-1}[\frac{5s+1}{s^2 - s - 12}] = 3e^{4t} + 2e_{3t}$$

EXAMPLE 14

$\mathcal{L}^{-1}[\frac{9s-8}{s^2-2s}]$

$$\frac{9s-8}{s^2-2s} \equiv \frac{A}{s} + \frac{A}{s-2}$$

$$9s-8 \equiv A(s-2) + B(s)$$

$s = 0 \qquad \therefore -8 = A(-2) + B(0) \rightarrow A = 4$

Put $s = 2$

$$\therefore 10 = A(0) + B(2) \rightarrow B = 5$$

$$F(t) = \mathcal{L}^{-1}[\frac{4}{s} + \frac{5}{s-2}]$$

$$= 4 + 5e^2 t$$

EXAMPLE 15

Express $f(s) = \frac{s^2-15s+41}{(s+2)(s-3)^2}$ in partial fractions and hence determine its inverse transform.

Solution

$$f(s) = \frac{s^2 - 15s + 41}{(s+2)(s-3)^2} \equiv \frac{A}{s+2} + \frac{B}{s-3} + \frac{C}{(s-3)^2}$$

$$s^2 - 15s + 41 \equiv A(s-3)^3 + B(s+2)(s-3) + C(s+2)$$

$$A = 3 \qquad B = -2 \qquad C = 1$$

$$\frac{s^2-15s+41}{(s+2)(s-3)^2} \equiv \frac{3}{s+2} - \frac{2}{s-3} + \frac{1}{(s-3)^2}$$

For $\mathcal{L}^{-1}[\frac{1}{(s-3)^2}]$ remember $\mathcal{L}^{-1}[\frac{1}{s^2}] = t$ and by Theorem 1, if $\mathcal{L}^{-1}[F(t)] = f(s)$ then $\mathcal{L}[e^{-at}F(t)] = f(s+a)$

.i.e. $\frac{1}{(s-3)^2}$ is like $\frac{1}{s^2}$ with s replaced by $(s-3)$, thus $a=-3$

$$\mathcal{L}^{-1}[\frac{1}{(s-3)^2}] = te^{3t}$$

$$\mathcal{L}^{-1}[\frac{s^2-15s+41}{(s+2)(s-3)^2}] = 3e^{-2t} - 2e^{3t} + te^3t$$

EXAMPLE 16

$\mathcal{L}^{-1}[\frac{4s^2-5s+6}{(s+1)(s^2+4}]$

SOLUTION

$$\frac{4s^2-5s+6}{(s+1)(s^2+4} \equiv \frac{A}{s+1} + \frac{Bs+C}{s^2+2}$$

$$A = 3 \qquad B = 1 \qquad C = -6$$

$$\mathcal{L}[F(t)] = \frac{3}{s+1} + \frac{s}{s^2+4} - \frac{6}{s^2+4}$$

$$F(t) = 3e^{-t} + \cos 2t - 3\sin 2t$$

EXERCISE

1. Find the inverse transforms of
   a. $\frac{1}{2s-3}$
   b. $\frac{5}{(s-4)^3}$
   c. $\frac{3s+4}{s^2+9}$
2. Express in Partial fractions
   a. $\frac{22s+16}{(s+1)(s-2)(s+3)}$
   b. $\frac{s^2-11s+16}{(s+1)(s-2)^2}$

3. Determine
   a. $\mathcal{L}^{-1}[\frac{4s^2-17s-24}{s(s+3)(s-4)}]$
   b. $\mathcal{L}^{-1}[\frac{5s^2-4s-7}{(s-3)(s^2+4)}]$

ANSWERS

1. a. $\frac{1}{2}e^{(\frac{3}{2}t)}$
   b. $5t^2e^{4t}$
   c. $3\cos 3t + \frac{4}{3}\sin 3t$
2. a. $\frac{1}{(s+1)} + \frac{4}{(s-2)} - \frac{5}{(s+3)}$
   b. $\frac{2}{(s+1)} - \frac{1}{s-2} - \frac{4}{(s-2)^2}$

3. a. $2 + 3e^{-3t} - e^{-4t}$
   b. $2e^{3t} + 3\cos 2t + \frac{5}{2}\sin 2t$

CONVOLUTION

Let $F(t)$ and $G(t)$ be piecewise-continuous functions on $[0, \infty]$ of exponential order. The function called convolution of $F(t)$ and $G(t)$, denoted by f,g is defined by:

$$(F * G)(t) = \int_0^t F(x)G(t-x)dx$$

THEOREM

Let F and G be piecewise continuous on$[0, \infty]$ of exponential order. Then,

$$\mathcal{L}[F * G] = \mathcal{L}[F]\mathcal{L}[G]$$

Equivalently,

$$\mathcal{L}^{-1}[\mathcal{L}[F]\mathcal{L}[G]] = F * G$$

EXAMPLE

Using convolution, find the inverse $F(t)$ of:

$f(s) = \frac{1}{(s^2+1)^2}$

SOLUTION

$$F(t) = \mathcal{L}^{-1}[f(s)] = \sin t * \sin t$$

$$= \int_0^t \sin\tau \sin(t-\tau) d\tau$$

$$= \frac{1}{2}\int_0^t -\cos t d\tau + \frac{1}{2}\int_0^t -\cos(2\tau - t) d\tau$$

$$= -\frac{1}{2} t\cos t + \frac{1}{2}\sin t$$

SOLUTION OF DIFFERENTIAL EQUATION BY LAPLACE TRANSFORMS

To solve a differential equation by Laplace transforms, we go through four distinct stages

a. Rewrite the equation in terms of Laplace transforms
b. Insert the given initial conditions
c. Rearrange the equation algebraically to give the transform of the solution
d. Determine the inverse transform to obtain the particular solution

LAPLACE TRANSFORM OF DERIVATIVES

$$\mathcal{L}\{y''\} = s^2\mathcal{L}\{y\}(s) - sy(s) - y'(0)$$

$$= s^2Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}(y') = s\mathcal{L}\{y\}(s) - y(0)$$

$$= sY(s) - y(0)$$

$$\mathcal{L}\{(y)(s)\} = Y(s)$$

$$\mathcal{L}\{y^k\} = s^kY(s) - s^{k-1}y(0) - \cdots - sy^{k-2}(0) - y^{k-1}(0)$$

EXAMPLE 17

Use Laplace transform to find the solution to the initial value problem $y'' - y = e^{2t}$ with $y(0) = a$ and $y'(0) = 1$

Taking the Laplace transform of the equation,

$$s^2 Y(s) - sy(0) - y'(0) - Y(s) = \frac{1}{s-2}$$

substituting the initial conditions and solving for Y we get

$$Y(s) = \frac{1}{(s+1)(s-2)}$$

Now,

$$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{(s-2)}$$

$1 = A(s-2) + B(s+1)$

substituting $s = -1$ gives $A = -\frac{1}{3}$

Similarly, substituting $s = 2$ reveals that $B = \frac{1}{3}$

Hence

$$Y(s) = \frac{1}{3}[\frac{1}{s-2} - \frac{1}{s+1}]$$

EXAMPLE 18

Use the Laplace transform to solve the initial value problem

$y'' - 2y' - 3y = 0 \qquad y(0) = 1 \quad y'(0) = 0.$

SOLUTION

$$\mathcal{L}\{y'' - 2y' - 3y\} = s^2Y(s) - sy(0) - y'(0) - 2[sY(s) - y(0)] - 3Y(s)$$

$$= (s^2 - 2s - 3)Y(s) - y(0)(s - 2) - y'(0)$$

substitution for $y(0) = 1$ and $y'(0) = 0$ gives

$$(s^2 - 2s - 3)Y(s) = s - 2$$

$$Y(s) = \frac{s-2}{s^2 - 2s - 3}$$

$$= \frac{\frac{1}{4}}{s-3} + \frac{\frac{3}{4}}{s+1}$$

Finding the Laplace inverse gives

$$y(t) = \frac{1}{4}e^{3t} + \frac{3}{4}e^{-t}$$

EXAMPLE 19

Find the solution to the initial value problem

$y^{iv} - y = 0 \qquad y(0) = 0 \qquad y'(0) = 1 \qquad y''(0) = 0 \qquad y'''(0) = 0$

SOLUTION

The Laplace transform the differential equation is

$$s^4 Y(s) - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0) - Y(s) = 0$$

Then using the initial conditions, we have

$$Y(s) = \frac{s^2}{s^4 - 1}$$

A partial expansion of $Y(s)$ is

$$Y(s) = \frac{As + B}{s^2 - 1} + \frac{Cs + D}{s^2 + 1}$$

And it follows that

$$(As + B)(s^2 + 1) + (Cs + D)(s^2 - 1) = s^2$$

Solving for the varies constants, we set $s = 1$,$s = -1$ and $s = 0$ respectively to obtain

$$Y(s) = \frac{\frac{1}{2}}{s^2 - 1} + \frac{\frac{1}{2}}{s^2 + 1}$$

Therefore the solution to the IVP is

$$y(t) = \frac{\sinh t + \sin t}{2}$$

EXAMPLE 20

Solve the differential equation (IVP)

$y'' - 3y' + 2y = e^{-t} \quad y(0) = 3 \quad y'(0) = 4$

SOLUTION

$$\mathcal{L}[y''] - 3\mathcal{L}[y'] + 2\mathcal{L}[y] = \mathcal{L}[e^{-t}] \cdots (1)$$

But

$$\mathcal{L}[y''] = s^2\mathcal{L}[y] - sy(0) - y'(0)$$

$$\mathcal{L}[y''] = s^2\mathcal{L}[y] - 3s - 4 \cdots (2)$$

and

$$\mathcal{L}[y'] = s\mathcal{L}[y] - y(0) = s\mathcal{L}[y] - 3 \cdots (3)$$

Put (3) and (2) into (1)

$$= (s^2\mathcal{L}[y] - 3s - 4) - 3(s\mathcal{L}[y] - 3) + 2\mathcal{L}[y] = \frac{1}{s+1}$$

$$(s^2 - 3s + 2)\mathcal{L}[y] = 3s - 5 + \frac{1}{s+1}$$

$$\mathcal{L}[y] = \frac{3s-5}{(s-2)(s-1)} + \frac{1}{(s-2)(s-1)(s+1)}$$

$$y = \mathcal{L}^{-1}[\frac{3s-5}{(s-2)(s-1)} + \frac{1}{(s-2)(s-1)(s+1)}]$$

$$\Rightarrow y = \mathcal{L}^{-1}[\frac{3s-5}{(s-2)(s-1)}] + \mathcal{L}^{-1}[\frac{1}{(s-2)(s-1)(s+1)}]$$

By partial fraction, we have:

$$\Rightarrow y = \mathcal{L}^{-1}[\frac{3s-5}{(s-2)(s-1)}] + \frac{1}{3}\mathcal{L}^{-1}[\frac{1}{s-2}] - \frac{1}{2}\mathcal{L}^{-1}[\frac{1}{s-1}] + \frac{1}{6}\mathcal{L}^{-1}[\frac{1}{s+1}]$$

$$\Rightarrow y = e^{2t} + 2e^{t} + \frac{1}{3}e^{2t} - \frac{1}{2}e^{t} + \frac{1}{6}e^{-t}$$

$$\Rightarrow y = \frac{4}{3}e^{2t} + \frac{3}{2}e^{t} + \frac{1}{6}e^{-t}$$

TRY THE FOLLOWING

1. $y^{'} + 3y = e^{2t} \quad y(0) = -1$
2. $y^{''} - 4y = e^{-t} \quad y(0) = -1 \quad y^{'}(0) = 0$

EXERCISE

1. If $Y(s) = \frac{v_0}{(s+c)^2-(c^2-w_0^2)}$. find $y(t)$
   Ans: $-\frac{v_0}{(2\sqrt{c^2-w_0^2})(e^{-(c+\sqrt{c^2-w_0^2})t}-e^{-(c-\sqrt{c^2-w_0^2})t})}$
2. Solve $y^{''} + 2y^{'} + 2y = \cos 2t$ with $y(0) = 0 \quad y^{'}(0) = 1$ using Laplace transform.

DIRAC'S DELTA FUNCTION

Definition

$\int_{-\infty}^{\infty} f(t)\delta(t-a)dt = f(a)$ and $\int_{-\infty}^{\infty} \delta(t-a)dt = 1$

Also $\int_{n}^{m} f(t)\delta(t-a)dt = f(a)$ and $\int_{n}^{m} \delta(t-a)dt$

provided in each case, that n¡a ¡m

In the Laplace transform form:

$$\mathcal{L}[f(t)\delta(t-a)] = f(a)e^{-as} \text{ and } \mathcal{L}[\delta(t-a)] = e^{-as}$$

NOTE : $\delta(t-a)$ is not a function in the ordinary sense.

EXAMPLES

1. $\mathcal{L}[\delta(t-3)] = e^{-3s}$
2. $\int_2^5 e^{-2t}\delta(t-3)dt = f(3) = e^{-2\times 3} = e^{-6}$
3. $\int_0^{\pi} \sin 5t \cdot \delta(t-\frac{\pi}{2})dt = f(\frac{\pi}{2}) = \sin\frac{5}{2}\pi = 1$

PARTIAL DERIVATIVES AND ITS INTEPRETATIONS

Let z=f(x,y) be a function of two independent variables x and y. The rate at which the function is changing at a point,(a,b). If we hold y fixed and allow x to vary or vice versa is called the **partial derivative of f(x,y)** with respect to x at (a,b) or the **partial derivative of f(x,y)** with respect to y at (a,b) and denoted by

$$\frac{\delta f}{\delta x}(a,b) = f_x(a,b) \text{ or } \frac{\delta f}{\delta y}(a,b) = f_y(a,b)$$

To compute $f_x(x, y)$, treat all the y's as constants (or numbers) and then differentiate the function with respect to x. Similarly, to compute $f_y(x, y)$, treat all the x's as constants (or numbers) and then differentiate the function with respect to y and differentiate the function with respect to y. The formal definitions of the two partial derivatives of f(x) above are given as

$$f_x(x, y) = \lim_{h \to 0} \frac{f(x+h,y)-f(x,y)}{h} \qquad f_y(x, y) = \lim_{h \to 0} \frac{f(x,y+h)-f(x,y)}{h}$$

The following are possible alternate notations for partial derivatives. Given the function z= f(x,y) the following are all equivalent notations.

$$f_x(x, y) = f_x = \frac{\delta f}{\delta x} = \frac{\delta}{\delta x}(f(x, y)) = D_x f$$

$$f_y(x, y) = f_y = \frac{\delta f}{\delta y} = \frac{\delta}{\delta y}(f(x, y)) = D_y f$$

EXAMPLE

(a) $f(x,y) = x^4 + 6\sqrt{y} - 10$

(b) $h(s,t) = t^2 \ln(s^2) + \frac{9}{t^3}$

(c) $w = x^2y - 10y^2z^2 + 43x - 7\tan(4y)$

SOLUTION

(a) $f_x(x,y) = 4x^3 \qquad f_y(x,y) = \frac{3}{\sqrt{y}}$

(b) $h_s(s,t) = \frac{\delta h}{\delta s} = t^7(\frac{2s}{s^2}) \qquad h_t(s,t) = \frac{\delta h}{\delta t} = 7t^6 \ln(s^2)$

(c) $\frac{\delta w}{\delta x} = 2xy + 43 \quad \frac{\delta w}{\delta y} = x^2 - 20yz^3 - 28\sec^2(4y) \quad \frac{\delta w}{\delta z} = -30y^2z^2$

INTERPRETATION OF PARTIAL DERIVATIVES

In this session, we will give two interpretations of Partial Derivatives. As with functions of single variables partial derivatives represent the rates of change of the function as the variable change. As we saw in the previous session, $f_x(x, y)$ represent the rate of change of the function f(x,y) as we change x and hold y fixed while $f_y(x, y)$ represent the rate of change of the function f(x,y) as we change y and hold x fixed.

EXAMPLE

Determine if $f(x,y) = \frac{x^2}{y^3}$ is increasing or decreasing at (2,5).

(a) if we allow x to vary and hold y fixed

(b) if we allow y to vary and hold x fixed

SOLUTION

(a) We need $f_x(x,y)$ and its value at the point.

$$f_x(x,y) = \frac{2x}{y^3} \Rightarrow f_x(2,5) = \frac{4}{125} > 0$$

Hence the function is increasing at (2,5) as we vary x and hold y .

(b) We need $f_y(x,y)$ and its value at the point.

$$f_y(x,y) = \frac{3x^2}{y^4} \Rightarrow f_x(2,5) = \frac{12}{625} < 0$$

Hence the function is decreasing at (2,5) as we vary y and hold x .

For a function of one variable $f(x), f'(a)$ represents the slope of the tangent line to y= f(x) at x=a. The partial derivatives $f_x(a, b)$ and $f_y(a, b)$ also represent the slopes of the trace of f(x,y) for the plane y=b and x=a at the point (a,b) respectively.

SYSTEM OF LINEAR DIFFERENTIAL EQUATIONS

INTRODUCTION

In real life situations quantities and their rate of change depend on more depend on more than one variable. For example, the rabbit population, though it may be represented by a single number, depends on the size of predator populations and the availability of food. In order to represent and study such complicated problems we need to use more than one dependent variable and more than one question.

Such problems lead naturally to a **System** of simultaneous Ordinary Differential Equations. Systems of differential equations are the tools to use. The differential equation are very much helpful in many areas of science . But most of interesting real life problems involve more than one unknown function.

Note that in the Lotka- Volterra system, the variable t is missing. This kind of system is called **autonomous system** and is written

$$\begin{cases} \frac{dy}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

We will usually denote the independent variable by t and the independent variables (the unknown functions of t) by $x_1, x_2, \cdots$, or $x, y, z, \cdots$

For instance, a system of two first order differential equations in the dependent variables x and y has the general form:

$$f\left(t, x, y, \frac{dx}{dt}, \frac{dy}{dt}\right) = 0$$
$$g\left(t, x, y, \frac{dx}{dt}, \frac{dy}{dt}\right) = 0$$

where f and g are given.

A solution of this system is a pair (x(t),y(t)) of functions of t that satisfy both equations identically over some real interval of values of t.

TYPES OF LINEAR SYSTEMS

DEFINITION 1

The general linear systems of two first-order differential equations in two unknown functions x and y are of the form:

$$a_1(t)x' + a_2(t)y' + a_3(t)x + a_4(t)y = f_1(t)$$

$$b_1(t)x' + b_2(t)y' + b_3(t)x + b_4(t)y = f_2(t)$$

where the primes denote derivatives with respect to the independent variable t.

i.e. $x' = \frac{dx}{dt}$ $\quad$ $y' = \frac{dy}{dt}$

NOTE

We shall be concerned with systems of this type that has a constant coefficients and the number of equations.

EXAMPLE

1. $x' + 3y' - 2x + y = t^2$
   $x' - 2y' - 3x - 5y = e^t$
2. $x' - 2y' + x - y = t^2 - 1$
   $x' + y' - x = t + 2$
3. (D+2)x-(D-1)y = 1
   (D+1)x-(D+2y) = t-1

The solution of the system is an ordered pair real functions(f,g) such that x=f(t),y=g(t) simultaneously satisfy both equations of the system on some real interval $a \leq t \leq b$

DEFINITION 2

The general linear systems of three first-order differential equations in three unknown functions x, y and z are of the form:

$$a_1(t)x' + a_2(t)y' + a_3(t)z' + a_4(t)x + a_5(t)y + a_6(t)z = f_1(t)$$

$$b_1(t)x' + b_2(t)y' + b_3(t)z' + b_3(t)x + b_4(t)y + b_6(t)z = f_2(t)$$

$$c_1(t)x' + c_2(t)y' + c_3(t)z' + c_3(t)x + c_4(t)y + c_6(t)z = f_3(t)$$

EXAMPLE

1. x'+z'-2x-5y+z = 1-t

   y' + 3x -5y -z = tan t

   $x' - y' + z' - 2z = e^t$

2. x' +y' -2z' +2x -3y +z =t

   $2x' - y' + 3z' + x + 4y - 5z \sin t$

   $x' + 2y' + z' - 3x + 2y - z = \cos t$

CONT. OF DEFN. 2

The solution of the system is an ordered triple real functions(f,g,h) such that x=f(t),y=g(t),z=h(t) simultaneously satisfy three equations of the system on some real interval $a \leq t \leq b$

Systems of the form (definition 1 and 2) contained only first derivatives and we now consider the basic linear system involving higher derivatives. This is the general linear system of two second order differential equations in two unknown functions x and y and is a system of the form:

$$a_1(t)x'' + a_2(t)y'' + a_3(t)x' + a_4(t)y' + a_5(t)x + a_6(t)y = f_1(t)$$

$$b_1(t)x'' + b_2(t)y'' + b_3(t)x' + b_4(t)y' + b_5(t)x + b_6(t)y = f_2(t)$$

An example is provided below with constant coefficient:

$$2x'' + 5y'' + 7x' + 3y' + 2y = 3t + 1$$

$$3x'' + 2y'' - 2y' + 4x + y = 0$$

For given fixed positive integers m and n, we could proceed in like manner, to exhibit other general linear systems of n,mth-order differential equations in n unknown functions and give examples each of such type of system .
Instead we proceed to introduce the standard type of linear system.
We now introduce standard type as a special case of the system (definition 1) of two first-order differential equations in two unknown functions x and y. We consider the special type of linear system, which is of the form:

$$x' = a_{11}(t)x + a_{12}(t)y + f_1(t)$$

$$y' = a_{21}(t)x + a_{22}(t)y + f_2(t)$$

This is called the **normal form** in the case of two linear differential equations in two unknown functions. The characteristic feature of such a system is apparent from the manner in which the derivatives appear in it. An example of such such a system with variable coefficients is

$$x' = t^2x + (t+1)y + t^3$$

$$y' = te^t + t^3y - e^t$$

while one with constant coefficients is

$$x' = 5x + 7y + t^2$$

$$y' = 2x - 3y - 2t$$

The normal form in the case of a linear system of three differential equations in three unknown functions x,y and z is

$$x' = a_{11}(t)x + a_{12}(t)y + a_{13}(t)z + f_1(t)$$

$$y' = a_{11}(t)x + a_{12}(t)y + a_{13}(t)z + f_2(t)$$

$$z' = a_{11}(t)x + a_{12}(t)y + a_{13}(t)z + f_3(t)$$

An example of such a system is the constant system

$$x' = 3x + 3y + +z + t$$

$$y' = 2x - 4y + 5z - t^2$$

$$z' = 4x + y - 3z + 2t + 1$$

The normal form in the general case of a linear system of n differential equations in n unknown functions $x_1, x_2, \cdots, x_n$ is

$$x' = a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n + f_1(t)$$

$$x_2' = a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n + f_2(t)$$

$$\vdots$$

$$x_n' = a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n + f_3(t)$$

An important fundamental property of a normal linear system is its relationship to a single nth-order linear differential equation in one unknown function. Specifically, consider the so-called normalized nth-order linear differential equation

$x^{(n)} + a_1(t)x^{(n-1)} + \cdots + a_{n-1}(t)x' + a_n(t)x = f(t) \cdots (\cdot)$

in the one unknown function x.

$$\left.\begin{aligned} x_1 &= x \\ x_2 &= x' \\ x_3 &= x'' \\ &\vdots \\ x_{n-1} &= x^{(n-2)} \\ x_n &= x(n-1) \end{aligned}\right\} \cdots (*)$$

From (*), we get

$x' = x_1^{'}, x'' = x_2^{'}, \cdots, x^{(n)} = x_n^{'} \cdots (**)$

Then using both * and **, the single nth-order equation can be transformed into

$$\left.\begin{aligned} x_1' &= x_2 \\ x_2' &= x_3 \\ &\vdots \\ x_{n-1}' &= x_n \\ x_n' &= -a(t)x_1 - a_{n-1}(t)x_2 - \cdots - a_1(t)x_n + f(t) \end{aligned}\right\} \cdots (***)$$

From (*), we get

$x' = x_1^{'}, x'' = x_2^{'}, \cdots, x^{(n)} = x_n^{'}$

which is a special case of the normal linear system of n equations in n unknown functions. Thus we see that single nth-order linear differential equation of the form $(\cdot)$ in one unknown function is indeed intimately related to a normal linear system of n first-order differential equation in n unknown functions.

EXAMPLE 1

Reduce the given equation to the normal form

$$\frac{d^2x}{dt^2} + e^t\frac{dx}{dt} + (t-1)x = 0$$

SOLUTION

Let $x = x_0 \implies \frac{dx}{dt} = \frac{dx_0}{dt} = x_1$

$x_1 \implies \frac{d^2x}{dt^2} = \frac{d^2x_0}{dt^2} = \frac{dx_1}{dt} = -e^t\frac{dx}{dt} - (t-1)x$

$\therefore$ the normal form is:

$$\left.\begin{aligned} \frac{dx_0}{dt} &= x_1 \\ \frac{dx_1}{dt} &= -e^t x_1 - (t-1)x_0 \end{aligned}\right\}$$

EXAMPLE 2

Reduce the equations to the normal form

$2\frac{dx}{dt} + \frac{dy}{dt} - 2x + 5y = 2t$

$\frac{dx}{dt} - 2\frac{dy}{dt} + x + 3t = 0$

SOLUTION

$$2\frac{dx}{dt} + \frac{dy}{dt} = 2x - 5y + 2t \cdots (i)$$

$$\frac{dx}{dt} - 2\frac{dy}{dt} = -x - 3t \cdots (ii)$$

Solving for $\frac{dx}{dt}$ by the method of elimination then $(i) \times 2 + (ii)$

$$\therefore 5\frac{dx}{dt} = 3x - 5y + t \implies \frac{dx}{dt} = 3x - y + \frac{1}{5}t$$

Also solving for $\frac{dy}{dt}$ then $(i) - (ii) \times 2$

$$\therefore 5\frac{dx}{dt} = 4x - 5y + 8t \implies \frac{dx}{dt} = \frac{4}{5}x - y + \frac{8}{5}t$$

Therefore we have the normal linear system as:

$$\left.\begin{array}{l} \frac{dx}{dt} = \frac{3}{5}x - y + \frac{1}{5}t \\ \frac{dy}{dt} = \frac{4}{5}x - y + \frac{8}{5}t \end{array}\right\}$$

SOLUTION TO A LINEAR SYSTEM OF DIFFERENTIAL EQUATIONS

The solution of the system:

$$\left.\begin{aligned} \frac{dx_1}{dt} &= f_1(t, x_1, x_2, \cdots, x_n) \\ \frac{dx_2}{dt} &= f_2(t, x_1, x_2, \cdots, x_n) \\ &\vdots \\ \frac{dx_n}{dt} &= f_n(t, x_1, x_2, \cdots, x_n) \end{aligned}\right\}$$

is the parametric vector $\begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$ on open real interval a¡t¡b for which the system is true and the vector x must be continuous on the open real interval.

METHODS OF SOLVING SLDE

The most elementary approach to System of Linear Differential Equations (SLDE) with constant coefficients involves the elimination of the dependent variables by appropriately combining pairs of equations.

The main methods we will consider in solving such systems are

(a) The differential operator

(b) The matrix method

THE DIFFERENTIAL OPERATOR METHOD

Recall that $D = \frac{d}{dt} \implies D^2 = \frac{d^2}{dt^2}$ and also $D^n = \frac{d^n}{dt^n}$

This method is more convenient in the case of manageable small systems' those consisting of no more than four equations.

EXAMPLE

Solve the system:

$(D-1)x - Dy = 2t+1$

$(2D+1)x + 2Dy = t$

SOLUTION

$$(D-1)x - Dy = 2t+1 \cdots (I)$$

$$(2D+1)x + 2Dy = t \cdots (II)$$

Subtracting twice (I) from (II), we have: $3x = -3t - 2$

Substituting $x = -t - \frac{2}{3}$ in (I),we obtain:

$Dy = 2t + 1 - (D-1)x = t + \frac{4}{3}$

$y = \frac{1}{2}t^2 + \frac{4}{3}t + c$

The complete solution: $x = -t - \frac{2}{3}, y = \frac{1}{2}t^2 + \frac{4}{3}t + c$